

Introduction

Psych 6810 Statistical Methods in Psychology I

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Day 1

Welcome

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Today

Carmen and the Syllabus

Grades

Lectures and Labs

Things to Do

Variables and Measurement

Carmen

Everything is under construction!

1. Syllabus
2. Resources
3. Schedule (flexible)
4. Misconduct
5. Feedback

Ungrading

1. Engagement is key
2. Final dossier
 - a. Exercise diary (≈ 4)
 - b. R notebooks (≈ 7)
 - c. Quizzes (≈ 4)
 - d. Extra material you've found on your own
3. Evaluation
 - a. 3-point scale
 - b. Midterm reflection and final dossier rubric
 - c. Self-grading and extra material
 - d. Final meeting during finals week

What is Allowed?

1. Group work (for exercises, with conditions)
2. Generative AI (for R notebooks, with conditions)
3. Quizzes are open everything except people

Misconduct

Lectures vs. Labs

1. Lectures: brief discussion of concepts with in-class exercises
2. Labs: review of material, Q&A, doing statistics with R

Things to Do Now

1. By Friday: Install R and RStudio
2. By next Monday: Find a dataset. Optimally, these data should be closely related to your first-year project.
 - a. Talk to your advisor
 - b. Search online (see resources in Foundations.I notebook)
 - c. Your dataset should comprise at least 5 variables including:
 - i. At least one dependent variable and four independent variables;
 - ii. At least two numeric variables and two categorical variables;
 - iii. At least one continuous variable; and
 - iv. At least 100 observations.

My GSS Data

The General Social Survey

<https://gssdataexplorer.norc.umd.edu/home>

Data from 1980 Respondents

- ▶ sibs: number of siblings, from 0 to 20 (OMG)
- ▶ age: age of respondent, from 18 to 89
- ▶ happy: rated level of happiness, where 1=very happy, 2=pretty happy, 3=not too happy
- ▶ health: condition of health, where 1=excellent, 2=good, 3=fair, 4=poor
- ▶ youngen: the response to a question about what the “younger generation” should be taught, where 1=taught by elders to do what is right and 2=taught to think for themselves even though elders may disapprove
- ▶ realrinc: the respondent's income in 1986 dollars, from \$755 to \$139,297 per year

Categorical vs. Numerical

(Qualitative/Nominal vs. Quantitative)

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Level of Measurement

Nominal, Ordinal, Interval, or Ratio:

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Continuous vs. Discrete

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Dependent vs. Independent

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- ▶ health: condition of health, where 1=excellent, 2=good, 3=fair, 4=poor
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